

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T27 · Simultaneous Equations

With Answers

Questions with full worked solutions

■ 56 pages

FAMILY MAP

Simultaneous Equations

F01	Solve a linear-quadratic system	11 Q 57 marks
F02	Eliminate between two linear equations	6 Q 18 marks
F03	Interpret a system graphically	6 Q 23 marks
F04	Substitute or model a linear system	1 Q 5 marks

22 Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

Worked Solution 13

Simultaneous equations with a quadratic substitution

UNIT 2
5 marks**Substitute**

$$y = x + 2 \Rightarrow x^2 + 3(x + 2) + (x + 2)^2 = 7.$$

Solve for x

$$2x^2 + 7x + 3 = 0 = (2x + 1)(x + 3) \Rightarrow x = -\frac{1}{2} \text{ or } -3.$$

Recover y

$$y = x + 2 \Rightarrow y = \frac{3}{2} \text{ or } -1.$$

Final answer

$$(x, y) = \left(-\frac{1}{2}, \frac{3}{2}\right) \text{ or } (-3, -1)$$

8 Solve the simultaneous equations

$$\begin{aligned}3x + 5y &= 8 \\4x + y &= -3.5\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 8 is 3 marks)

Worked Solution 23

Simultaneous linear equations

UNIT 2

3 marks

Eliminate x

$$5(4x + y) = -17.5 \Rightarrow 20x + 5y = -17.5; \quad (20x + 5y) - (3x + 5y) = -17.5 - 8.$$

Solve

$$17x = -25.5 \Rightarrow x = -1.5, \quad y = -3.5 - 4(-1.5) = 2.5.$$

Final answer

$$x = -1.5, \quad y = 2.5$$

Question 16

4MA1-2H Q10

ORIGINAL QUESTION

3 marks

10 Solve the simultaneous equations

$$\begin{aligned}6x + 4y &= 1 \\ 3x + 5y &= 8\end{aligned}$$

Show clear algebraic working.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

Worked Solution 16

Simultaneous equations

MODULAR UNIT 2

3 marks

Eliminate

$$2(3x + 5y = 8) \Rightarrow 6x + 10y = 16; \quad 6y = 15.$$

Finish

$$y = 2.5, \quad x = -1.5.$$

Final answer

$$x = -1.5, \quad y = 2.5$$

Question 25

4MA1-2H Q22

ORIGINAL QUESTION

5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.....,) (.....,)

Worked Solution 25

Line-curve intersections

MODULAR UNIT 2

5 marks

Substitute

$$y = 5 - x : \quad 2x^2 + 3(5 - x)^2 = 210.$$

Solve

$$x^2 - 6x - 27 = 0 \Rightarrow x = 9, -3.$$

Coordinates

$$y = -4, 8.$$

Final answer

$$(9, -4), (-3, 8)$$

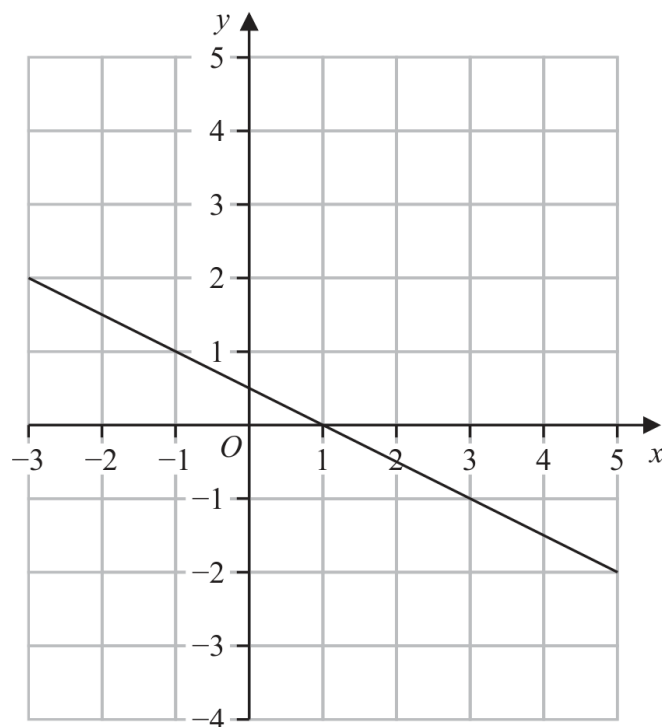
Question 29

4MA1-2H Q14 whole

ORIGINAL QUESTION UNIT 2

3 marks

14 Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$

$$2y + x = 1$$

$x = \dots\dots\dots$

$y = \dots\dots\dots$

Worked solution 29

Graphical solution of simultaneous equations

MODULAR UNIT 2

3 marks

whole – Graphical solution of simultaneous equations

Draw the second line

$$y - x - 2 = 0 \Rightarrow y = x + 2; \text{ draw it through } (-2, 0), (0, 2), (2, 4).$$

Read the intersection

The two drawn lines meet at $(-1, 1)$.**Final answer**

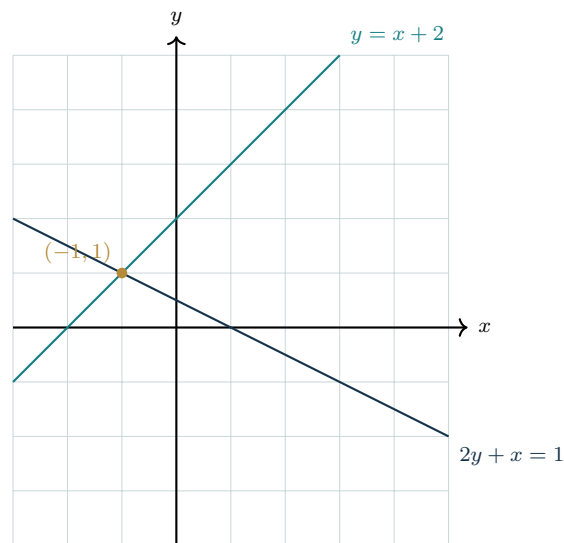
$$x = -1, y = 1$$

Completed diagram 29

Graphical solution of simultaneous equations

MODULAR UNIT 2

3 marks

Completed solution diagram: Graphical intersection

Question 34

4MA1-2H Q21 whole

ORIGINAL QUESTION UNIT 2

5 marks

- 21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(..... ,)

(..... ,)

Worked solution 34

Intersections of a line and an ellipse-type curve

MODULAR UNIT 2

5 marks

whole – Intersections of a line and an ellipse-type curve

Substitute the line equation

$$x = 5 - 2y \Rightarrow (5 - 2y)^2 + 3y^2 = 13.$$

Form a quadratic

$$7y^2 - 20y + 12 = 0 = (7y - 6)(y - 2).$$

Solve for y

$$y = \frac{6}{7} \text{ or } y = 2.$$

Recover x

$$y = \frac{6}{7} \Rightarrow x = \frac{23}{7}; \quad y = 2 \Rightarrow x = 1.$$

State both points

$$(1, 2) \text{ and } \left(\frac{23}{7}, \frac{6}{7}\right).$$

Final answer $(1, 2) \text{ and } (23/7, 6/7)$

Question 09

4MA1-1H Q9

ORIGINAL QUESTION

3 marks

- 9 Solve the simultaneous equations

$$2x + 9y = 14.5$$

$$7x + 3y = 8$$

Show clear algebraic working.

$x =$

$y =$

Worked Solution 09

Linear simultaneous equations

MODULAR UNIT 2

3 marks

Eliminate y

$$3(7x + 3y = 8) \Rightarrow 21x + 9y = 24.$$

Subtract

$$19x = 9.5 \Rightarrow x = 0.5.$$

Substitute

$$7(0.5) + 3y = 8 \Rightarrow y = 1.5.$$

Final answer

$$x = 0.5, \quad y = 1.5$$

Question 28

4MA1-2H Q21

ORIGINAL QUESTION

5 marks

21 Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

Worked Solution 28

Linear-quadratic simultaneous equations

MODULAR UNIT 2

5 marks

Substitute

$$x = 3y - 1 \Rightarrow 2(3y - 1)^2 + 3y^2 = 11.$$

Form the quadratic

$$21y^2 - 12y - 9 = 0 \Rightarrow 7y^2 - 4y - 3 = 0.$$

Solve

$$(7y + 3)(y - 1) = 0 \Rightarrow y = 1 \text{ or } y = -\frac{3}{7}.$$

Find x

$$y = 1 \Rightarrow x = 2; \quad y = -\frac{3}{7} \Rightarrow x = -\frac{16}{7}.$$

Final answer

$$(x, y) = (2, 1) \text{ or } \left(-\frac{16}{7}, -\frac{3}{7}\right)$$

Original question 34

4MA1-2H Q22 M01

MODULAR UNIT 2

5 marks

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

.....

Worked solution 34

Solve simultaneous equations

MODULAR UNIT 2

5 marks

M01 – Solve simultaneous equations**Substitute $y=x+2$**

$$x^2 + y^2 + y = 3 \Rightarrow x^2 + (x + 2)^2 + (x + 2) = 3.$$

Form the quadratic

$$2x^2 + 5x + 3 = 0 = (2x + 3)(x + 1).$$

Find x-values

$$x = -3/2 \text{ or } x = -1.$$

Find y-values

$$y = x + 2 \Rightarrow y = 1/2 \text{ or } y = 1.$$

Pair the solutions

$$(-3/2, 1/2) \text{ and } (-1, 1).$$

Final answer

$$\left(-\frac{3}{2}, \frac{1}{2}\right) \text{ and } (-1, 1)$$

Original question 24

4MA1-2H Q10 Q10

ORIGINAL QUESTION**3 marks****10** Solve the simultaneous equations

$$7x + 3y = 3$$

$$3x - y = 7$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

Worked solution 24

Chapter: Simultaneous Equations

MODULAR UNIT 2

3 marks

Q10 – Chapter: Simultaneous Equations

Family: Eliminate between two linear equations • Variant: Direct linear elimination

MethodMultiply $3x - y = 7$ by 3 to get $9x - 3y = 21$.*Why: Create opposite y -coefficients for elimination.***Method**Add to $7x + 3y = 3$: $16x = 24$, so $x = 3/2$.*Why: Eliminate y and solve for x .***Method**Then $y = 3x - 7 = -5/2$.*Why: Substitute x into either original equation.***Final answer****x:** $x = 1.5$ **Final answer****y:** $y = -2.5$

Original question 12

4MA1-1H Q19

ORIGINAL QUESTION

5 marks

19 Solve the simultaneous equations

$$\begin{aligned} 3x^2 + y^2 - xy &= 5 \\ y &= 2x - 3 \end{aligned}$$

Show clear algebraic working.

Worked solution 12

Chapter: Simultaneous Equations

MODULAR UNIT 2

5 marks

Q19 – Chapter: Simultaneous Equations

Family: Solve a linear-quadratic system • Variant: Line with mixed quadratic relation

MethodSubstitute $y=2x-3$ into $3x^2+y^2-xy=5$ to obtain $5x^2-9x+4=0$.*Why: Use the given linear relation to form one quadratic.***Method** $(5x-4)(x-1)=0$, so $x=0.8$ or $x=1$.*Why: Factorise the quadratic.***Method**For $x=0.8$, $y=2(0.8)-3=-1.4$.*Why: Substitute the first x -value.***Method**For $x=1$, $y=2(1)-3=-1$.*Why: Substitute the second x -value.***Method**The solutions are $(0.8, -1.4)$ and $(1, -1)$.*Why: State both ordered pairs.***Final answer****solutions:** $(0.8, -1.4)$ and $(1, -1)$

Original question 24

4MA1-2H Q10

ORIGINAL QUESTION**3 marks****10** Solve the simultaneous equations

$$3x + 5y = 3.1$$

$$6x + 3y = 3.75$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

Worked solution 24

Chapter: Simultaneous Equations

MODULAR UNIT 2

3 marks

Q10 – Chapter: Simultaneous Equations

Family: Eliminate between two linear equations • Variant: Decimal or fractional simultaneous coefficients

MethodMultiply the first equation by 2: $6x + 10y = 6.2$.*Why: Make the x-coefficients equal.***Method**Subtract $6x + 3y = 3.75$ to get $7y = 2.45$, so $y = 0.35$.*Why: Eliminate x and solve for y.***Method**Substitute $y = 0.35$: $6x + 1.05 = 3.75$, hence $x = 0.45$.*Why: Use either original equation.***Final answer****x,y:** $x = 0.45$, $y = 0.35$

- 22** The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B .

Find the coordinates of A and B .
Show clear algebraic working.

(..... ,)

(..... ,)

Worked solution

January 2020 | Question 22 | 5 marks

Family: Solve a linear-quadratic system | Variant: Multiple solution-pair validation

METHOD / WORKING

1. *Substitute* $y = x + 2$ into $x^2 + y^2 - 2y = 24$.

2. *Obtain* $x^2 + x - 12 = 0$.

3. Factor: $(x+4)(x-3)=0$.

4. $x=-4$ gives $y=-2$; $x=3$ gives $y=5$.

5. State both points.

FINAL ANSWER

(-4,-2) and (3,5)

19 Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$

$$x + 2y - 1 = 0$$

Show clear algebraic working.

.....

Worked solution

January 2021 | Question 19 | 5 marks

Family: Solve a linear-quadratic system • Variant: Multiple solution-pair validation

Method / working

1. Substitute the linear equation into the quadratic to obtain a three-term quadratic.
2. Solve it to get the two x-values $x=-2$ and $x=-7$.
3. Substitute $x=-2$ to obtain $y=3/2$.
4. Substitute $x=-7$ to obtain $y=4$.
5. State both paired solutions.

Final answer: $(x,y)=(-2,3/2)$ or $(-7,4)$

8 Solve the simultaneous equations

$$5a + 2c = 10$$

$$2a - 4c = 7$$

Show clear algebraic working.

$$a = \dots\dots\dots$$

$$c = \dots\dots\dots$$

Worked solution

June 2021 | Question 8 | 3 marks

Family: Eliminate between two linear equations • Variant: Scaled linear elimination

Method / working

1. Multiply the first equation by 2 and combine with $2a - 4c = 7$ to eliminate c .
2. Solve for $a = 2.25 = 9/4$.
3. Substitute back to obtain $c = -0.625 = -5/8$.

Final answer: $a = 2.25$, $c = -0.625$

19 Solve the simultaneous equations

$$\begin{aligned}y &= 3 - 2x \\ x^2 + y^2 &= 18\end{aligned}$$

Show clear algebraic working.

Worked solution

June 2021 | Question 19 | 5 marks

Family: Solve a linear-quadratic system | Variant: Line with circle or sum-of-squares curve

METHOD / WORKING

1. *Substitute* $y = 3 - 2x$ into $x^2 + y^2 = 18$.2. *Simplify* to $5x^2 - 12x - 9 = 0$.3. Factor: $(5x+3)(x-3)=0$, giving $x=-.6$ or 3 .4. Substitute x values to get $y=4.2$ or -3 .

5. State the two ordered pairs clearly.

FINAL ANSWER

(-0.6, 4.2) and (3, -3)

- 6 Alison buys 5 apples and 3 pears for a total cost of \$1.96
Greg buys 3 apples and 2 pears for a total cost of \$1.22

Michael buys 10 apples and 10 pears.

Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

\$.....

Worked solution

November 2021 | Question 6 | 5 marks

Family: Substitute or model a linear system | Variant: Contextual simultaneous price model

METHOD / WORKING

1. Let a be the cost of one apple and p be the cost of one pear.

2. $5a+3p=1.96$

3. $3a+2p=1.22$

4. Multiply the first equation by 2:

5. $10a+6p=3.92$

6. Multiply the second equation by 3:

7. $9a+6p=3.66$

8. Subtract:

9. $a=0.26$

10. Then

11. $3(0.26)+2p=1.22$

12. $2p=0.44$

13. $p=0.22$

14. Michael buys 10 apples and 10 pears:

15. $10(0.26)+10(0.22)=4.80$

16. Answer: \$4.80.

FINAL ANSWER

\\$4.80.

- 19 The straight line **L** has equation $x - y = 3$
The curve **C** has equation $3x^2 - y^2 + xy = 9$

L and **C** intersect at the points P and Q .

Find the coordinates of the midpoint of PQ .
Show clear algebraic working.

(.....,)

Worked solution

November 2021 | Question 19 | 6 marks

Family: Solve a linear-quadratic system | Variant: Line with mixed quadratic relation

METHOD / WORKING

1. Line L is
2. $x - y = 3$
3. So
4. $y = x - 3$
5. The curve is
6. $3x - y + y^2 = 9$
7. Substitute $y = x - 3$:
8. $3x - (x - 3) + (x - 3)^2 = 9$
9. $2x + 3 + x^2 - 6x + 9 = 9$
10. $x^2 - 4x + 3 = 0$
11. $(x - 1)(x - 3) = 0$
12. So $x = 1$ or $x = 3$.
13. If $x = 1$, then $y = -2$. If $x = 3$, then $y = 0$.
14. The midpoint of $(1, -2)$ and $(3, 0)$ is
15. $(\frac{1+3}{2}, \frac{-2+0}{2}) = (2, -1)$
16. Answer: $(2, -1)$.

FINAL ANSWER $(2, -1)$.

12 Solve the simultaneous equations

$$7x - 2y = 34$$

$$3x + 5y = -3$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

Answer reference | May-Nov 2020 | Question 12

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

12	Elimination E.g. $21x - 6y = 102$ $21x + 35y = -21$ $(-41y = 123)$ or $35x - 10y = 170$ $6x + 10y = -6$ $(41x = 164)$	Substitution E.g. $3\left(\frac{34+2y}{7}\right) + 5y = -3$ or $3x + 5\left(\frac{7x-34}{2}\right) = -3$ or $7\left(\frac{-3-5y}{3}\right) - 2y = 34$ or $7x - 2\left(\frac{-3-3x}{5}\right) = 34$	4	M1 for a correct method to eliminate x or y : coefficients of x or y the same and correct operation to eliminate selected variable (condone 1 arithmetical error) or for correctly writing x or y in terms of the other variable and correctly substituting
				A1 dep on M1 for $x = 4$ or $y = -3$
	E.g. $7x - 2 \times -3 = 34$			M1 dep on M1 for substitution of found variable or repeating the steps in first M1 for the second variable
		$x = 4$ $y = -3$		A1 cao A correct answer without working scores no marks
Total 4 marks				

Worked solution

May-Nov 2020 | Question 12 | 4 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Solution

$$7x - 2y = 34$$

$$3x + 5y = -3$$

Multiply the first equation by 5:

$$35x - 10y = 170$$

Multiply the second equation by 2:

$$6x + 10y = -6$$

Add:

$$41x = 164$$

$$x = 4$$

Substitute into $3x + 5y = -3$:

$$12 + 5y = -3$$

$$y = -3$$

Answer: $x=4$, $y=-3$.

16 Solve the simultaneous equations

$$\begin{aligned}3xy - y^2 &= 8 \\ x - 2y &= 1\end{aligned}$$

Show clear algebraic working.

(Total for Question 16 is 5 marks)

Answer reference | May-Nov 2020 | Question 16

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

16	$3y(2y + 1) - y^2 = 8$ or $x = \frac{8 + y^2}{3y} \rightarrow \frac{8 + y^2}{3y} - 2y = 1$ or $-3xy - y^2 = 8$ $3xy - 3y \times 2y = 3y \times 1$ oe	$3x\left(\frac{x-1}{2}\right) - \left(\frac{x-1}{2}\right)^2 = 8$ oe			M1 correct first step eg substitution by eg $x = 1 + 2y$ or $y = \frac{x-1}{2}$ to get an equation in a single variable or writing 2 nd equation with x the subject and substituting into 1 st or multiplying 2 nd equation by $3y$ and subtracting from 1 st oe
	eg $5y^2 + 3y - 8 (= 0)$	eg $5x^2 - 4x - 33 (= 0)$			A1 for a correct simplified quadratic
	$(5y + 8)(y - 1) (= 0)$ or $\frac{-3 \pm \sqrt{3^2 - 4 \times 5 \times (-8)}}{2 \times 5}$	$(5x + 11)(x - 3) (= 0)$ or $\frac{4 \pm \sqrt{(-4)^2 - 4 \times 5 \times (-33)}}{2 \times 5}$			M1ft dep on M1 for solving their 3 term quadratic equation using any correct method (allow one sign error and some simplification – allow as far as $\frac{-3 \pm \sqrt{9 + 160}}{10}$) or if factorising, allow brackets which expanded give 2 out of 3 terms correct)
	$y = -\frac{8}{5}$ and $y = 1$ (both)	$x = -\frac{11}{5}$ and $x = 3$ (both)			A1 dep on first M1
			$x = -\frac{11}{5}, y = -\frac{8}{5}$ $x = 3, y = 1$	5	A1 oe dep on first M1 Must be paired correctly
					Total 5 marks

Worked solution

May-Nov 2020 | Question 16 | 5 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Official final mark-scheme pages are embedded immediately after this question.

7 Solve the simultaneous equations

$$\begin{aligned}5x + 4y &= -2 \\ 2x - y &= 4.4\end{aligned}$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 7 is 3 marks)

Answer reference | January 2023 | Question 7

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Question	Working	Answer	Mark	Notes
7	eg $5x + 4y = -2$ $+ 8x - 4y = 17.6$ $(13x = 15.6)$ eg $[x = \frac{4.4 + y}{2}]$ oe $5\left(\frac{4.4 + y}{2}\right) + 4y = -2$ oe	eg $10x + 8y = -4$ $- 10x - 5y = 22$ $(13y = -26)$ eg $[y = 2x - 4.4]$ oe $5x + 4(2x - 4.4) = -2$ oe	3	M1 multiplication of one or both equation(s) with correct operation selected (allow one arithmetic error) (if + or - is not shown then assume it is the operation that at least 2 of the 3 terms have been calculated for) or correct rearrangement of one equation with substitution into second
	eg $5 \times "1.2" + 4y = -2$ or $2 \times "1.2" - y = 4.4$	eg $5x + 2 \times "-2" = 4.4$ or $2x - "-2" = 4.4$		M1 (dep on previous M1 but not on a correct first value) correct method to find second unknown – this could be a correct substitution into one of the equations given or calculated or starting again with the same style of working as for the first method mark
	Working required	$x = 1.2$ $y = -2$		A1 oe eg $x = \frac{6}{5}$ for both solutions dependent on first M1
Total 3 marks				

Worked solution

January 2023 | Question 7 | 3 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Official final mark-scheme pages are embedded immediately after this question.

20 Solve the simultaneous equations

$$\begin{aligned}y &= 7 - 2x \\ x^2 + y^2 &= 34\end{aligned}$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

Answer reference | January 2023 | Question 20

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

20	$x^2 + (7 - 2x)^2 = 34$	$\left(\frac{7-y}{2}\right)^2 + y^2 = 34$	5	M1	substitution of linear equation into quadratic
	$5x^2 - 28x + 15 [= 0]$ oe	$5y^2 - 14y - 87 [= 0]$ oe		M1	dep on previous M1 for multiplying out and collecting terms, forming a three term quadratic in any form of $ax^2 + bx + c (= 0)$ where at least 2 coefficients (a or b or c) are correct and all are non-zero
	$(5x - 3)(x - 5) [= 0]$ or $\frac{-(-28) \pm \sqrt{(-28)^2 - 4 \times 5 \times 15}}{2 \times 5}$ or $5\left(x - \frac{28}{10}\right)^2 - \frac{784}{100} + 15 = 0$ oe or $x = 0.6$ and $x = 5$ (allow incorrect labels for x/y)	$(5y - 29)(y + 3) [= 0]$ or $\frac{-(-14) \pm \sqrt{(-14)^2 - 4 \times 5 \times (-87)}}{2 \times 5}$ or $5\left(y - \frac{14}{10}\right)^2 - \frac{196}{100} - 87 = 0$ oe or $y = 5.8$ and $y = -3$ (allow incorrect labels for x/y)		M1ft	dep on M1 for solving <i>their</i> 3 term quadratic equation using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error and some simplification – allow as far as $\frac{28 \pm \sqrt{784 - 300}}{10}$ or $\frac{14 \pm \sqrt{196 + 1740}}{10}$) (if completing the square allow as far as shown) or correct values for x or correct values for y dep on correct quadratic
	eg $y = 7 - 2 \times 5$ and $y = 7 - 2 \times 0.6$ (correct labels for x/y)	eg $5.8 = 7 - 2x$ and $-3 = 7 - 2x$ (correct labels for x/y)		M1ft	dep on previous M1 for substituting their 2 found values of x or y in a suitable equation or correct values for the other variable
	<i>Working must be shown</i>		$x = 0.6, y = 5.8$ $x = 5, y = -3$	A1	dep on M1 and the correct quadratic (allow coordinates) must be paired correctly

Answer reference | January 2023 | Question 20

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

				Total 5 marks
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Worked solution

January 2023 | Question 20 | 5 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Official final mark-scheme pages are embedded immediately after this question.

- 28** The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are $(0.2, 2.6)$

Work out the coordinates of B
Show your working clearly.



(..... ,)

(Total for Question 28 is 6 marks)

Worked solution

November 2025 | Question 28 | 6 marks

Family: Solve a linear-quadratic system | Variant: Line with circle or sum-of-squares curve

METHOD / WORKING

Worked solution

November 2025 | Question 28 | 6 marks

Family: Solve a linear-quadratic system | Variant: Line with circle or sum-of-squares curve

Method

The line is

$$y=3x+e$$

Point A=(0.2,2.6) lies on the line.

$$2.6=3(0.2)+e$$

$$2.6=0.6+e$$

$$e=2$$

So

$$y=3x+2$$

The curve is

$$x^2 + y^2 = d - 11x$$

Use point A=(0.2,2.6):

$$0.2^2 + 2.6^2 = d - 11(0.2)$$

$$0.04 + 6.76 = d - 2.2$$

$$d=9$$

Now substitute $y=3x+2$ into the curve:

$$x^2 + (3x + 2)^2 = 9 - 11x$$

$$x^2 + 9x^2 + 12x + 4 = 9 - 11x$$

$$10x^2 + 23x - 5 = 0$$

One root is $0.2 = \frac{1}{5}$.

The product of the roots is

$$\frac{-5}{10} = -\frac{1}{2}$$

So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

Then

$$y=3(-\frac{5}{2})+2$$

$$y=-\frac{11}{2}$$

Answer:
$$B = (-\frac{5}{2}, -\frac{11}{2})$$

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER

2 Solve the simultaneous equations

$$5x + y = 11$$

$$3x - y = 9$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 2 is 3 marks)

Worked solution

November 2025 | Question 2 | 3 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Solution

$$5x+y=11$$

$$3x-y=9$$

Add the equations:

$$8x=20$$

$$x=2.5$$

Substitute into $5x+y=11$:

$$5(2.5)+y=11$$

$$y=-1.5$$

Answer: $x=2.5$, $y=-1.5$.